

Week 5

Learning from Random Samples

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Speaker notes

No notes on this slide.

Motivating Example

Roasting Coffee: Introduction

Speaker notes

- This is the set shot
- Coffee is one of life's innumerable joys. It is also the best of them.
- There are flavors, complexity, bitterness, sweetness – and caffeine.
- I've been roasting coffee at home, for several years, I'll take you on a little tour. Don't worry, there's a reason.

Roasting Coffee: Field Trip

Speaker notes

- Introduce (a) green coffee (and its range of size and dryness); (b) the roaster; (c) the temperature scope; (d) the young gun; and (e) the tryer.
- Introduce the tryer, then show trying before drop.
- Within each of these “trys” I’ve got a sample that has been jumbled up for 10 minutes. I’ve drawn about 30 beans, each from the same distribution.
- Ask students: In casual terms, would you say that the beans that I pull out are “random”?
- I sure would!
- But, for our statistics to work, we need something that is different, and more specific than “random” – we will need independent and identically distributed.
- The jumbling of these beans means that each bean’s draw is independent of each other bean in the barrel.

Statistics, Parameters, and Estimators

Introduction to Properties of Estimators

Estimating Characteristics of Populations

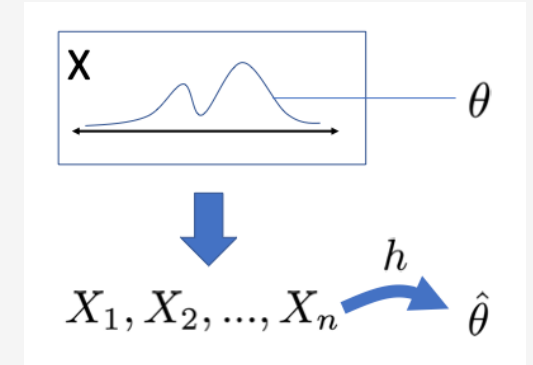
- Data scientists are not given the joint density distribution.
- One of our principle tasks is to produce *estimates* about the *universe* given only the limited information that we get from a sample of data.

Estimating Characteristics of Populations (cont.)

i Parameters, Statistics, and Estimators

- A **parameter**, θ , is some summary of the joint distribution.
- A **statistic** is some function that maps data from $\mathbb{R}^n \rightarrow \mathbb{R}$.
- An **estimator**, $\hat{\theta}$, is some statistic of the data that we use to produce a guess for θ .

Speaker notes



- We can't observe the joint distribution directly, so we try to make guesses about it.
- An estimator could be any function of the statistic. Here, we'll pass the data through function h and get estimator

$\overline{X}$ is an Estimator

We refer to statistics that are “good” guesses for a population parameter as **estimators**.

- $\theta = \mathbb{E}[X]$ is a parameter.
- $T_{(n)} =$ choose the 30th percentile is a statistic
- $\hat{\theta} = \overline{X}$ is an estimator for θ .

- **Consistency**
- **Bias**
- **Efficiency**

Introducing Estimators

In this section, you will learn about the properties that make estimators good guesses for a population parameter.

Properties of estimators

- **Consistency:** As sample size grows, does the estimator converge in probability to the true value?
- **Bias and unbiasedness :** Is the estimator systematically too low or too high?
- **Efficiency:** Does an estimator have relatively *large or small* sampling variance, standard error, and MSE?

Reading: Core Estimation Theory

Reading: Core Estimation Theory

Note: This is a READING CALL, just placing it here for organization.

Read pages 102–105, stopping at 3.2.3, Variance Estimators.

Desirable Properties of Estimators

Desirable Properties of Estimators

Throughout, we refer to θ as a population *feature* or *parameter* .

- θ has a fixed, true value, but this value is not directly observable.
- Without ground truth, how can we know if our estimator is doing its job?

Speaker notes

- Test-driven development, engineering, analogy – software teams imagine every scenario possible, write a testing suite – and conduct tests before releasing. That’s how they know it “works”. Because we don’t know
- Reasoning and evaluating these properties is one of the reasons that we have invested in probability theory!

Desirable Properties of Estimators: Unbiasedness

Bias of an estimator

The bias of an estimator is the expected difference between the *true* population value and the estimator. - The *bias* of $\hat{\theta}$ is $\mathbb{E}[\hat{\theta}] - \theta$. - If $\mathbb{E}[\hat{\theta}] = \theta$, then there is no bias, and the estimator is *unbiased*.

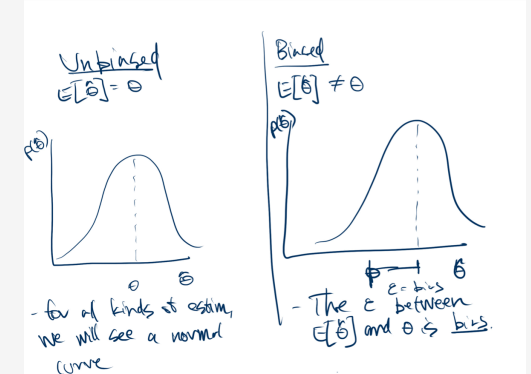
Desirable Properties of Estimators: Unbiasedness (cont.)

If $\hat{\theta}$ is unbiased, does that mean that, for any sample taken, the estimate produced by $\hat{\theta} = \theta$?

- $\hat{\theta}$ as an *estimator* is a random variable.
- The value that it takes on given a sample is the *estimate*.

Speaker notes

- Draw an unbiased estimator!
- As a RV,
- The
- Draw a biased estimator!



Desirable Properties of Estimators: Unbiasedness

(cont.)

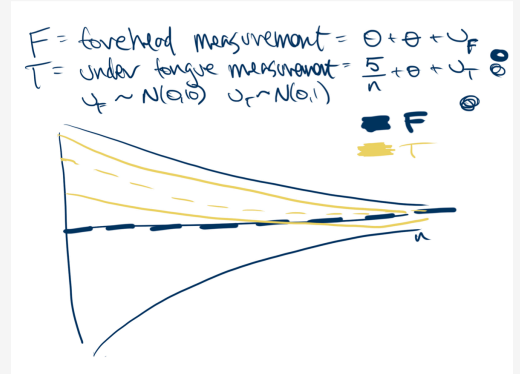
Desirable Properties of Estimators: Efficiency

i Mean Squared Error of an Estimator

- The MSE of an estimator $\hat{\theta}$ is $E[(\hat{\theta} - \theta)^2] = V[\hat{\theta}] + (E[\hat{\theta}] - \theta)^2$

Speaker notes

- Suppose you are taking temperature, and have two methods: (1) back of hand; (2) under-tongue. Which would you prefer?



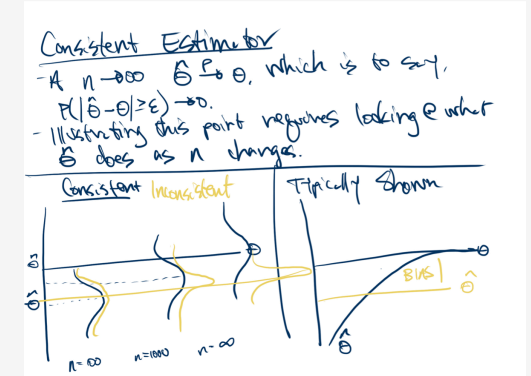
Desirable Properties of Estimators: Efficiency

Desirable Properties of Estimators: Consistency

i Consistency

An estimator $\hat{\theta}$ is consistent for θ if $\hat{\theta} \xrightarrow{p} \theta$.

[Note: If an estimator isn't consistent, then many times we would say it isn't worth using.]



Desirable Properties of Estimators: Consistency

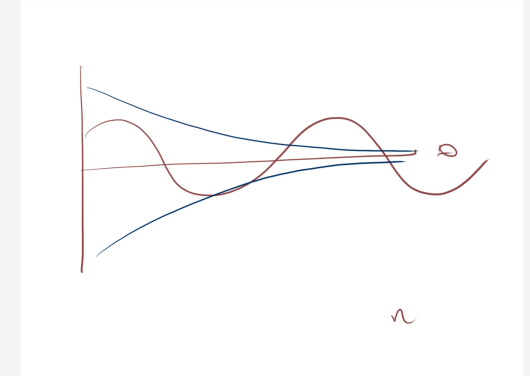
Applying Properties of Estimators

Evaluating Estimators

Which do you prefer?

Speaker notes

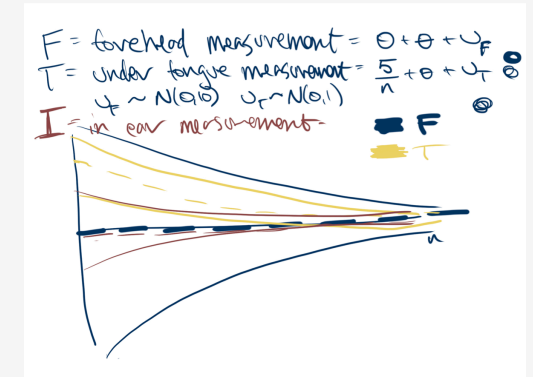
- Draw two estimators: (1) sine in n – wavy; (2) typically converging



Evaluating Estimators (cont.)

Speaker notes

- Draw two estimators
- Make both be consistent, one that is relatively more efficient.



Desirable Properties of Estimators: Review

- An *inconsistent* estimator is of little use.
- A *more efficient* estimator is preferable to a *less efficient* estimator, all else equal.
- An *unbiased* estimator is preferable to a *biased* estimator, all else equal.

Speaker notes

- For inconsistent estimators, perhaps there is some level of
- More efficient estimators are preferable, all else equal. But, in a business context there

Random Sampling

Reading: Independent and Identically Distributed

Reading: Independent and Identically Distributed

- Read Section 3.0 and 3.1 in *Foundations of Agnostic Statistics* (pages 91–96 in our copy of the book.) - Work to place the formal math definition into plain understanding. We will discuss this when you come back.
- Notice how μ and σ^2 are concepts that are immediately familiar, but they now represent population parameters.

Speaker notes

- The mathematical formalism in
- *Definition 3.1.2 – the Finite Population Mass Function* shows how we could move from talking about actual people to random variables (but keep in mind that we will almost always assume an “infinite superpopulation”.)
- Notice how

Independent and Identically Distributed

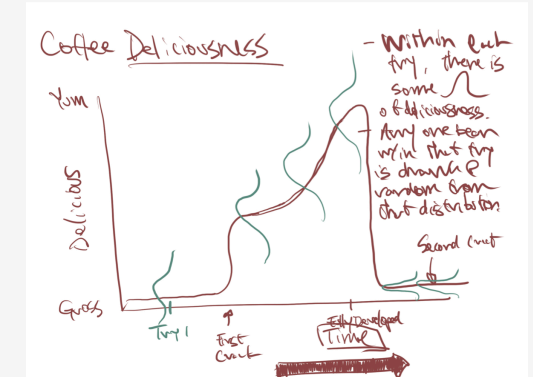
Independent and Identically Distributed

Definition: independent and identically distributed (IID)

- Undertake some process an arbitrary number of times—call it *sampling*—to create a value from a phenomenon.
- If each instance of sampling draws from the same probability distribution, then we say the collection of values are **identically distributed**.
- If none of the instances of sampling provide information about other instances of sampling, then we say the collection of values are **independent**.

Independent and Identically Distributed (cont.)

Speaker notes



- Each of the sets of
- Problem 1: We have a finite sample in this roaster – there are 10,243 beans in it. Paul counted! Ha! After we draw and look at one, there are only 10,242 left to draw, and we've changed the distribution of the beans by taking one out. So, technically, we have a dependent sampling process.
- But, subsequent ``trys'' are not independent. When I know how roasted each bean in try (3) is, I know that each bean in try (4) is going to be more roasted.
- Problem 2: The beans only get

Making Assumptions

- What is a statistical assumption?
- When are they important?
- When can I violate them?
- What do I do if they are violated?

Speaker notes

- Up until this point, all of our work has been axiomatic.
- We began with a principle, we looked at rules, and we observed what fell out.
- Now we're stepping into the fun, but clumsy world that forces us to put these alabaster axioms to work.

Making Predictions

- When data are independent and identically distributed (IID), it is possible to learn desirable things from a sample: -
Accurate characterizations of the probability distribution function and values
- Reliable characterizations of certainty and uncertainty
- We can learn about unseen *population parameters* from a sample and then use our domain knowledge to evaluate whether these population parameters apply to some circumstance.

Speaker notes

- As data scientists, we might sometimes possess the entire universe of data and want to characterize it – “What is the sentiment of all the employees in
- Much more frequently we’re interested in something else – making a statement about a population from limited data.
- What sets statistics apart is that it attempts to characterize how certain we are about these predictions.
- Without IID, we might be incorrectly confident – leading to unforeseen events. Or, we might learn something about a population that isn’t informative of our target.

What Is a Population?

U.S. Senator fundraising

There are 100 U.S. Senators.

- One could reason about a finite population with 100 elements. `\note[item]{Then, we \textit{know} the whole population, and there is nothing to estimate.}`

- Or, one could assume a continuous distribution for fundraising.

Speaker notes

- When we model data as draws from a random variable
- X

Learnosity: Is This IID?

Is This IID?

Note: This is a learnosity activity, just placing it here for organization.

For each of these, would you say that this sample is distributed IID? What do you understand about the process that brings you to your conclusion?

- Coffee - **Goal:** Understand how roasted a batch of coffee is.
- **Sampling process:** Dip into a drum to pull a set of 30 beans.
- Conduct a draft - **Goal:** Randomly select people for military service.
- **Sampling process:** Draw balls from an urn with a day of the year. Everybody born on that day goes to war.

Is This IID? (cont.)

- Teach a computer to *really* read - **Goal:** Teach a computer to understand theme and plot (the work of School of Information professor David Bamman).
- **Sampling process:** Feed a neural network each word (in each sentence [in each paragraph (in each chapter)]) of the English language literary canon.
- Any others? - **Goal:**
- **Sampling process:**

Reading: Sample Statistics

Reading: Sample Statistics

- Read pages 96–98.
- Stop before theorem 3.2.5.

Sample Statistics

Sample Statistics

Sample statistics are:

- Functions that are applied to samples of random variables
- *Themselves* random variables

Speaker notes

- At this point, we don't have any guarantee about what the sample statistics will look like under repeated sampling. If you've taken a stats course before, though, you'll recognize that we're building towards sampling distribution of these statistics.

Conduct Sampling

Conduct Sampling

Note: This is a learnosity activity, just placing it here for organization.

The goal is that this will solidify the understanding that students have from the reading and will move us forward into a conversation about expected value and the sample variance of the sample average.

- Students will be provided with data and starter code that produces a population.
- From this population, they will have sliders they can pull that will increase or decrease the number of samples that they take, the number of draws per sample, and the underlying population variation.

The Sample Mean as an Estimator

The Sample Mean

Definition: Sample Mean

For IID random variables, $X_1, \dots, X_n$, the `_sample mean_` is

```
\[
\overline{X} = \frac{1}{n} \sum_{i=1}^n X_i
\]
```

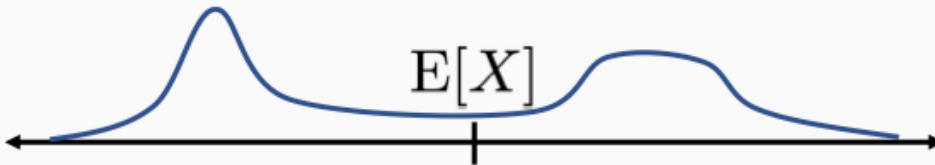
- The sample mean is an *estimator* for $\mathbb{E}[X]$.

Speaker notes

- Let's dive into our first example of an estimator: the sample mean.
- I'm sure you've taken thousands of means in your life, but you might not have thought of the sample mean as an estimator.
- First, what kind of object is this? It's a statistic. It's a function of a sample, which is made up of random variables, so it is a random variable. This is really our statistical model that describes how a sample mean you write down in your analysis gets there.
- As statisticians, we're not really interested in the sample mean for its own sake. We're interested in the population mean. We're using the sample mean, because it's an estimator for the population mean.

The Sample Mean is an Estimator

1. Population Distribution



2. Sample



3. Sampling Distribution of the Statistic



Speaker notes

- Here's a picture to clarify how the model works.
- There are three important objects to keep in mind.
- First, we have the population distribution, we don't know what it is, but it's fixed, and it has a fixed population mean.
- What happens when we run the model? nature reaches in and chooses datapoints. I'll make a sample.
- From this sample, we compute the sample mean. Does it equal the population mean? of course not! well, in some cases we could get lucky, but generally no.
- what kind of object is the sample mean? it's a random variable. if you could rewind time, we'd get a different mean.
- I'm not saying we can get two means, this is just a thought experiment.
- because the sample mean is a random variable, it has a sampling distribution. I'll draw it

Evaluating the Sample Mean as an Estimator

First Questions:

1. Is the sample mean biased?
2. How efficient is the sample mean?

Speaker notes

- Once you start thinking about the sample mean as an estimator, the next step is to think about whether it's a good estimator. Are there any better estimators?
- What are the properties that we want our estimators to have?
- Let's start in this segment with these two.

Unbiasedness of the Sample Mean

Theorem: The Sample Mean is Unbiased

For IID random variables, $X_1, \dots, X_n$, the sample mean $\overline{X}$ is an unbiased estimator for the population mean $E[X]$.

Speaker notes

- $E[\overline{X}] = E\left[\frac{1}{n}(X_1 + X_2 + \dots + X_n)\right]$
- $= \frac{1}{n}(E[X_1] + \dots + E[X_n]) = E[X]$

Efficiency of the Sample Mean

Theorem: Sampling Variance of the Sample Mean

For IID random variables, $X_1, \dots, X_n$, with population variance, $V[X]$, the variance of the sample mean is

$$V[\overline{X}] = \frac{V[X]}{n}$$

Speaker notes

- $V[\overline{X}] = V[\frac{1}{n}(X_1 + X_2 + \dots + X_n)]$
- $= \frac{1}{n^2}(V[X_1] + \dots + V[X_n]) = \frac{V[X]}{n}$
- Is this a good variance? It really depends on the population distribution. Depending on what the population looks like, there could be more efficient estimators. But at least the dependency on
- Soon, we'll see that we can get a guarantee that the sample mean is actually close to the target.

Learnosity: Sampling Variance of the Sample Mean

Note: This is a learnosity activity, just placing it here for organization. - If you increase the sample size, does the sampling variance of the sample mean increase, decrease, or stay the same? - At what rate does this change? (Options include faster than the data, at the same rate at the data, slower than the data.) - If you increase the sample size, does the population variance, $V[X]$, change? (Hint: It is a population parameter.)

Consistency and the Continuous Mapping Theorem

Consistency

“If you can’t get it right as n goes to infinity, you shouldn’t be in this business.”

How can we formalize “right as n goes to infinity?”

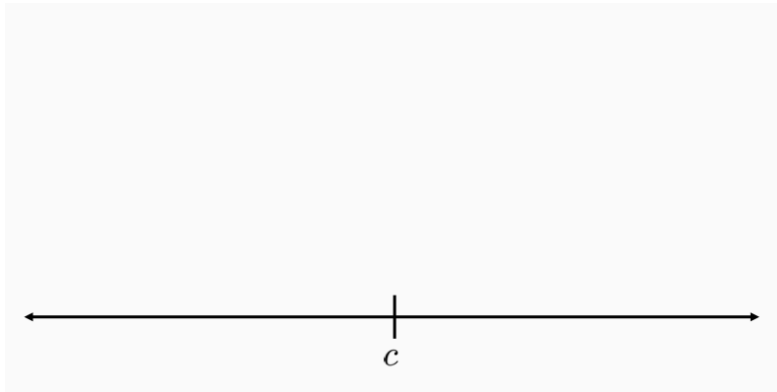
- Estimators may converge at different rates.
- Deterministic guarantees are not possible.

Speaker notes

- It’s no exaggeration to say that consistency is the most important property that an estimator can have.
- So what exactly does it mean?
- A quote from the Nobel-prize winning economist Clive Granger.
- We want to be correct as the sample size gets bigger and bigger - how can we write that into a mathematical property?
- it’s not obvious for at least two reasons.
- First, different estimators can converge at different rates. it may take
- Second, there’s no deterministic guarantee. There’s no law of physics that prevents a coin from landing heads over and over forever. all we can say is that the probability of that gets smaller and smaller as the number of flips increases.

Intuition for Convergence in Probability

- Let $T_{(1)}$ be the statistic with 1 datapoint.
- Let $T_{(2)}$ be the statistic with 2 datapoints.
- Let $T_{(3)}$ be the statistic with 3 datapoints.



Speaker notes

- Here's a stylized story to motivate the definition of consistency.
- Let's start with a sequence of random variables
- Let
- You tell me you want to converge on the target. So I say, well give me some parameters, give me a window around the target that is acceptable. also, 100
- I guarantee, that eventually in this sequence, there is a T that is in that window with 90
- Is the guarantee not strong enough? feel free to increase it: let's say 99
- Is the window actually too wide? make it narrower!...

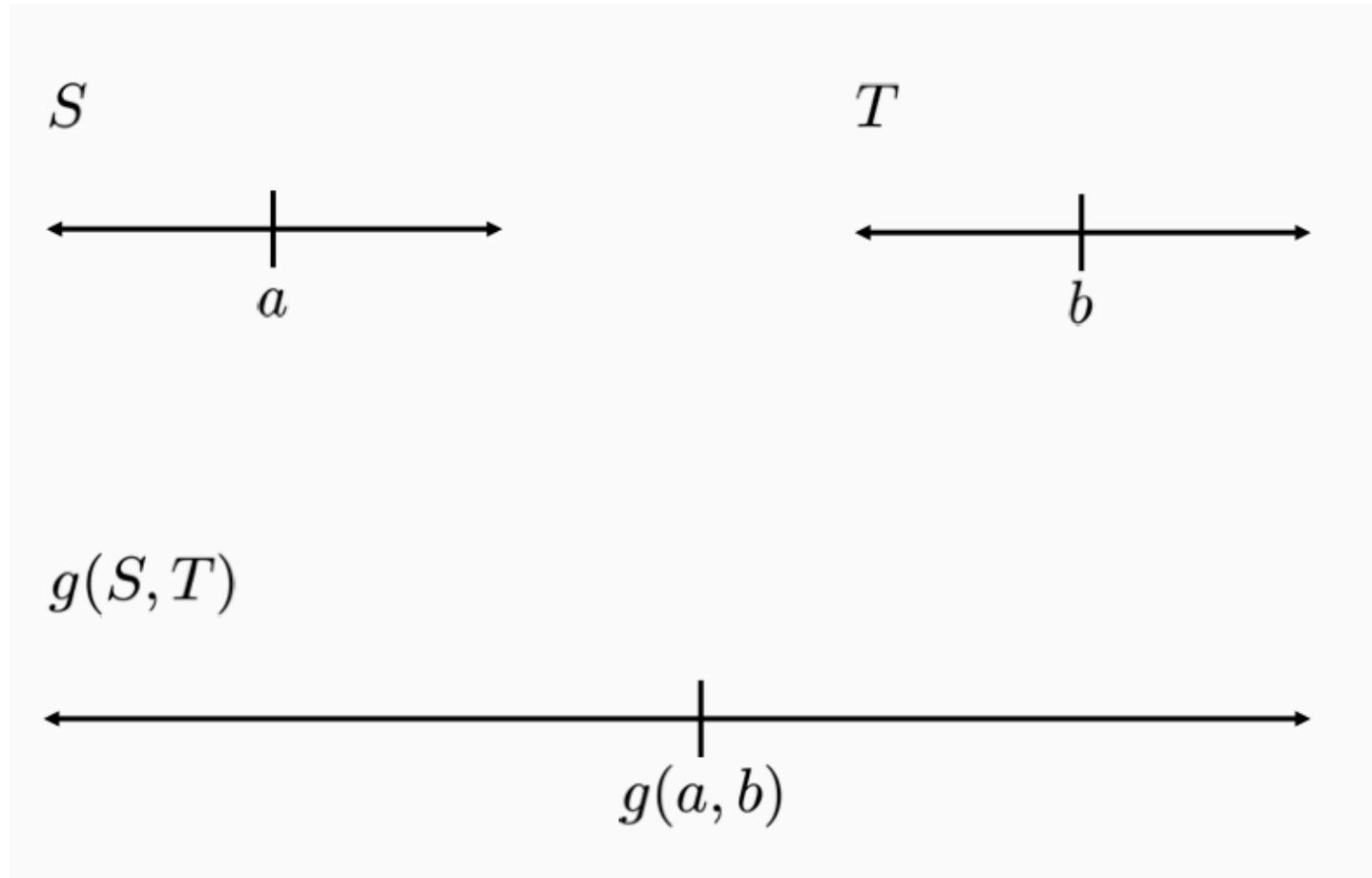
Convergence in Probability

i Definition: Convergence in Probability

Let $(T_{(1)}, T_{(2)}, T_{(3)}, \dots)$ be a sequence of random variables and let $c \in \mathbb{R}$.
 $T_{(n)}$ converges in probability to c if for all $\epsilon > 0$,
 $\lim_{n \rightarrow \infty} P\Big[T_{(n)} \in (c-\epsilon, c+\epsilon) \Big] = 1$
 We write this as $T_{(n)} \xrightarrow{p} c$.

- An estimator $\hat{\theta}$ is *consistent* for θ , if $\hat{\theta} \xrightarrow{p} \theta$.

Continuous Mapping Intuition



Speaker notes

- Consistency is really important. We're going to design a lot of estimators and we'll want to prove that they are consistent.
- There are different strategies you can use to prove that estimators are consistent. The book likes to use a very powerful result about the plug in principle, but you have to know a lot more math to apply it rigorously.
- We prefer to use a famous theorem called the continuous mapping theorem. It allows you to prove convergence in probability when you combine estimators together.
- Here's a picture to show what we mean. we have two estimators S and T that each converge to some target. We have a function g that takes a, b to this point. As long as g is consistent, it turns out that $g(S, T)$ is also consistent.

The Continuous Mapping Theorem

Let $(S_{(1)}, S_{(2)}, S_{(3)}, \dots)$ and $(T_{(1)}, T_{(2)}, T_{(3)}, \dots)$ be two sequences of random variables. Let $g: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a continuous function. If $S_{(n)} \xrightarrow{p} a \in \mathbb{R}$ and $T_{(n)} \xrightarrow{p} b \in \mathbb{R}$, then $g(S_{(n)}, T_{(n)}) \xrightarrow{p} g(a, b)$

Speaker notes

- Given
- Want
- Equivalent to: given probability
- By cont. of
- by convergence of S and T,
- let
- This implies

Reading: Weak Law of Large Numbers

Reading: Weak Law of Large Numbers

Read page 100, beginning at theorem 3.2.8, through the end of page 102.

Weak Law of Large Numbers

The Weak Law of Large Numbers

i Theorem: The Weak Law of Large Numbers

Let $(X_1, X_2, X_3, \dots)$ be a sequence of i.i.d. random variables with finite variance. Let $\overline{X}_{(n)} = \frac{1}{n} \sum_{i=1}^n X_i$. Then $\overline{X}_{(n)} \xrightarrow{p} \mathbb{E}[X]$

- **Equivalently:** The sample mean is *consistent* for the population mean.

Speaker notes

- We have all the pieces in place to present one of the most classic results in statistics.
- The WLLN states that as sample size increases, the sample mean converges in probability to the population mean.
- Or much more simply, the sample mean is consistent.

Consequences of WLLN

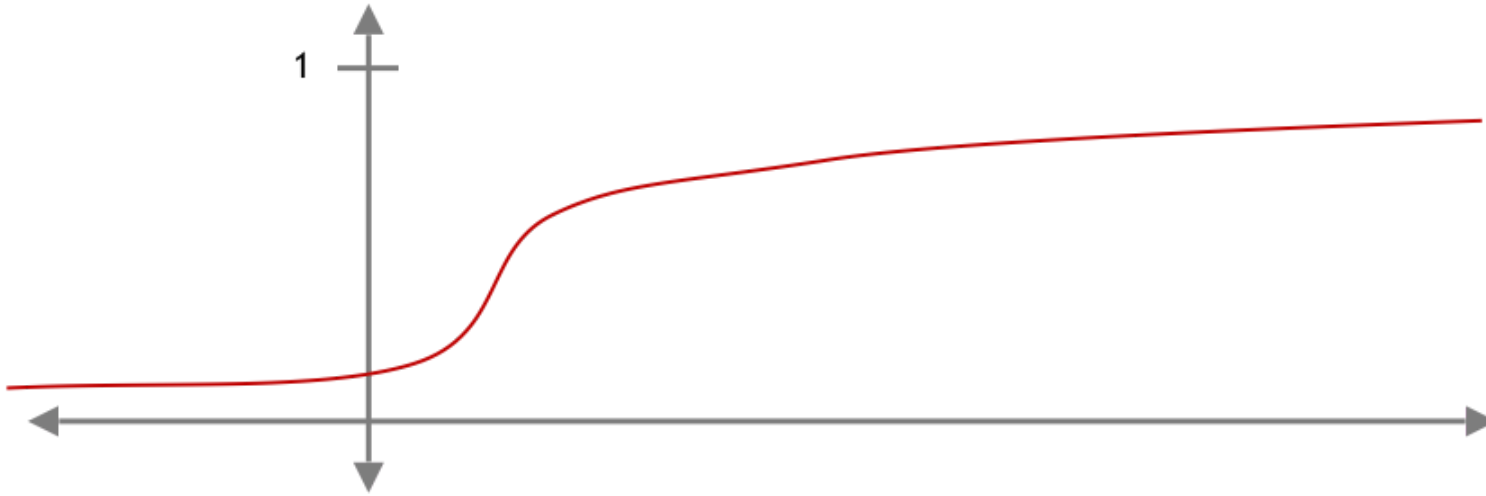
The sample mean is consistent. - The error converges in probability to zero. - The sample mean is accurate, as long as we can make n large enough.

As a building block to study other estimators. - Combine WLLN with Continuous Mapping Theorem.

Speaker notes

- What do we take away from the WLLN?
- First, we learned that the sample mean is great. On top of being unbiased, we also have consistency.
- Whatever the variable is: wavelengths of cosmic rays, or how cranky people in Berkeley are. You never know the true population mean, but if you crank up n enough, you'll can be accurate to any level.
- Now, honestly, you already knew the sample mean was great. you don't need more reasons to use the mean. And we can write down more precise statements about the convergence of the sample mean. So why is the WLLN so important?
- But is a second reason WLLN is important. it is a building block we can use to study other estimators.

Applying the WLLN



Objective: Estimate the cdf F at a point x .

Idea: Use empirical cdf

Speaker notes

- $F(x) = P(X \leq x)$
- $T_n(x) = \frac{1}{n} \sum I(X_i \leq x)$
- $T_n(x) \xrightarrow{P} \mathbb{E}[I(X \leq x)] = P(X \leq x) = F(x)$
- So we can consistently estimate the CDF at any point.
- There are stronger convergence results for the cdf, so we don't use the WLLN version too much
- But this is an example of how WLLN can help us with other estimators. we'll use it again in the future.

Proof of the Weak Law of Large Numbers

Proof of the Weak Law of Large Numbers

Given random variable X and $\epsilon > 0$.

Proof of the Weak Law of Large Numbers

Given random variable X and $\epsilon > 0$.

$$\text{Let } D = |\bar{X} - \mathbb{E}[X]| \quad \mathbb{V}[\bar{X}] = \mathbb{E}[D^2] = \mathbb{V}[X]/n$$

$$\mathbb{V}[\bar{X}] = \mathbb{E}[D^2] = \mathbb{E}[D^2 | D < \epsilon]P(D < \epsilon) + \mathbb{E}[D^2 | D \geq \epsilon]P(D \geq \epsilon)$$

$$\geq 0 + \epsilon^2 P(D \geq \epsilon) P(D \geq \epsilon) \leq \frac{\mathbb{V}[X]}{\epsilon^2 n} \xrightarrow{p} 0$$

Speaker notes

- The key to this proof is we have to get from a small variance to a statement about deviations.
- Why can't we be outside this window? Let's draw the probability distribution.
- The point is that if there's some probability outside the window, that represents deviations that are large, and that has the effect of keeping variance up. that can't happen.

Simulating the WLLN

Note: This is a learnosity activity, we're just including it here for organization.

Students will work through the notebook called

Reading: Estimating Population Variance

- **Note:** This is a **READING CALL**, just placing it here for organization.

Reading: Estimating Population Variance

- Read pages 105–108, stopping before the beginning of the next section.
- You are going to use *the plug-in principle* , which is a general approach for designing estimators in a sample.

Estimating Population Variance

Why Estimate Variance?



- How much coffee do students drink on average?
- How much does coffee intake differ from student to student?

Speaker notes

- So far, we've talked about estimating the population mean.
- The mean is really important, but there are other quantities that we might want to estimate.
- Let's consider the variance - usually the second-most important feature you can learn.
- If you estimate mean coffee intake, that's helpful, but just one number, nothing about the shape of the distribution.
- IF you also know variance, you start to have at least some information about shape. That can really improve your understanding.

Applying the Plug-In Principle

Population Variance: $V[X] = E[X^2] - E[X]^2$

The Plug-In Variance Estimator

Given a sample $\text{bs}X = (X_1, X_2, \dots, X_n)$, the *plug-in variance estimator* is,

$$\tilde{V}(\text{bs}X) = \overline{X^2} - \overline{X}^2$$

Speaker notes

- How do we come up with an estimator for variance?
- We use what we call the plug-in principle. All that means is that we just replace population quantities with sample analogues.
- In this case, we can write the variance as the difference of two expectations.
- We just replace each expectation with a sample mean.

Consistency of the Plug-In Estimator

i Theorem: Consistency of the Plug-In Variance Estimator

Let $(X_1, X_2, X_3, \dots)$ be a sequence of i.i.d. random variables with finite variance $V[X]$.
Then $\tilde{V}(\text{bs}X) = \overline{X^2} - \overline{X}^2$ is consistent for $V[X]$.

Speaker notes

- Proof:
- $\overline{X^2} \xrightarrow{p} E[X^2]$
- Let
- $g(\overline{X^2}, \overline{X}) = \tilde{V}(\text{bs}X)$
- CMT

Bias of the Plug-In Variance Estimator

Let $(X_1, X_2, X_3, \dots)$ be a sequence of i.i.d. random variables with finite variance $V[X]$. Then

$$E[\tilde{V}(X)] = \frac{n-1}{n} V[X]$$

Speaker notes

- What about bias? Unfortunately, the plug-in estimator is biased. That's pretty common, the plug-in principle often gives us consistent estimators, but doesn't help with bias.
- You can compute the expectation of the estimator, and you can see that it's almost what you want.
- There's just an extra factor of
- If you think about it, if we knew the true population mean, we could measure deviations from it to get an unbiased estimator. But we have the sample mean, and the sample mean gets pulled towards where the most data is, pushing the estimate down.

A Better Variance Estimator

i The Unbiased Variance Estimator

Given a sample $\text{bs}X = (X_1, X_2, \dots, X_n)$, the *unbiased sample variance estimator* is,

$$\hat{V}(\text{bs}X) = \frac{n}{n-1} \left(\overline{X^2} - \overline{X}^2 \right)$$

Speaker notes

- So the plug-in estimator is biased, but we know exactly what the bias is so we can fix it.
- We just adjust it by adding a correction factor. It's called a degrees of freedom correction.
- This makes the estimator unbiased, and it's still consistent, because this fraction goes to 1.
- This estimator has really good properties, and it's the standard formula you should remember.

Standard Errors

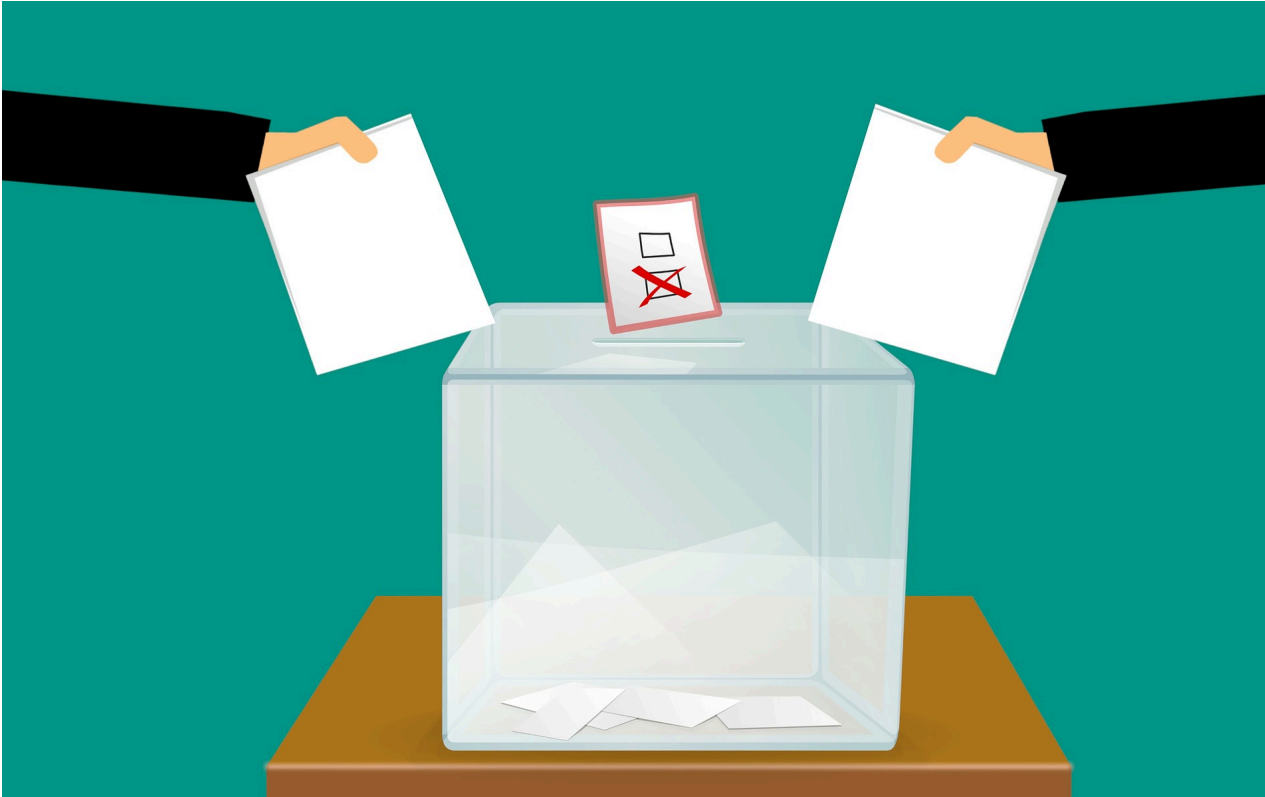
standout

Point Estimate $\Leftrightarrow$ Uncertainty

Speaker notes

- We've been talking about ways to generate a point estimate for a parameter.
- But the point estimate is really only half the story.
- That's because the point estimate is not the same as the true parameter.
- If all you have is the point estimate, and no other way to judge how far off it might be, it may not be very valuable at all.
- If you're just messing around with data for fun, you may not care, but when the stakes get high, you realize you really need a way to capture that uncertainty.

Importance of Uncertainty



Estimate: Our candidate has 52 % support.

Importance of Uncertainty



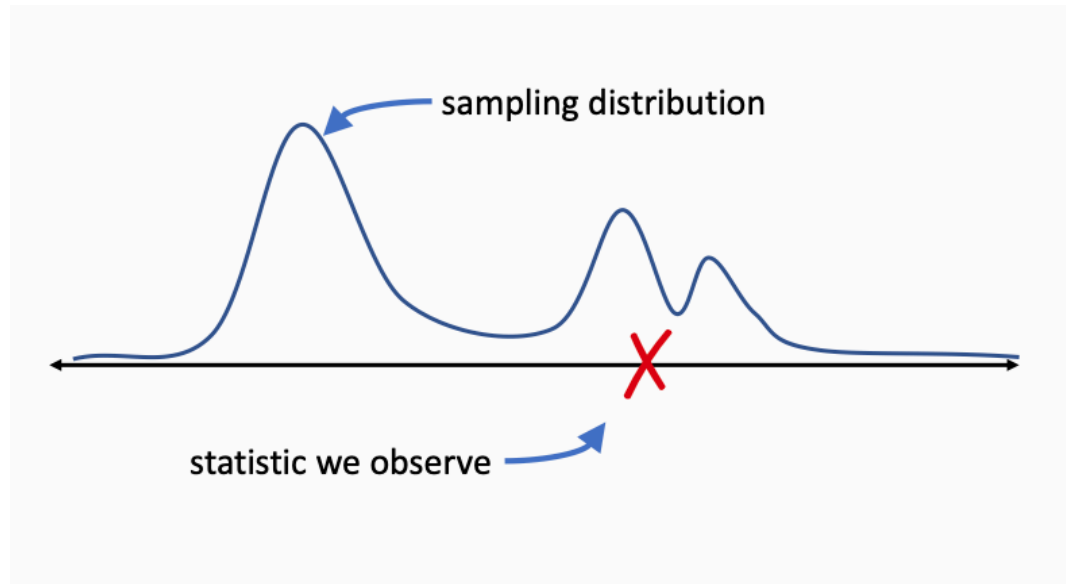
Estimate: The maximum dose of the medication is 26mg.

Importance of Uncertainty



Estimate: The angle of the tower is 90 degrees.

The Sampling Distribution of the Statistic



Standard Error: (Estimated) standard deviation of the sampling distribution

Speaker notes

- Remember that a statistic IS a random variable
- You get a single number. But the model says that there are other values that are possible.
- There is an entire probability distribution over possible values.
- If we could see this distribution, we could understand how representative our one value is.
- We could gauge how different it might be from the true parameter.
- We can't see the sampling distribution, but if i.i.d. holds, we can still learn things about it.
- A good first question: how spread out is it? standard error = standard deviation of the sampling dist.
- You can think of the standard error as estimating, if you could run your study over and over, how far apart the values would be on average. that's an important indicator of uncertainty

Reporting standard errors

1. The mean number of mushrooms per pizza was 13.2 (SE 3.6).
2. The time for mice to navigate the maze was 35 ± 2 seconds.
- 3.

Speaker notes

- Really, whenever you report a statistic, you always want to include standard errors.
- unless you have another method of expressing uncertainty.
- There are a couple of ways to include them...

Standard Errors, The Sample Variance, and Standard

Deviation

Many Measures of Dispersion

 Sample Variance

 Sample Standard Deviation

 Sampling Variance of the Sample Mean

 Standard Error of the Sample Mean

Speaker notes

- It can seem like we're just making a salad with all these concepts!
- I mean, I am a Berkeley vegetarian, but even I don't like variance salads.
- The
- The
- The
- Finally, the

Motivating The Central Limit Theorem

Capturing Uncertainty

Need tools to capture how far off our estimate may be.

- Standard Error: the (estimated) standard deviation of the sampling distribution of the estimator.
- But a standard error is just one number...

Speaker notes

- We've talked about how important uncertainty is.
- A point estimate by itself isn't all that useful. we need a way to capture how far off it might be
- One answer to that is the standard error, which is an estimate for the standard deviation of the sampling distribution of a statistic.
- But the standard error doesn't capture everything about the sampling distribution.
- It's just one number. it can give you a general idea, but can't help you make precise statements about what parameter values are plausible.

Uncertainty Example

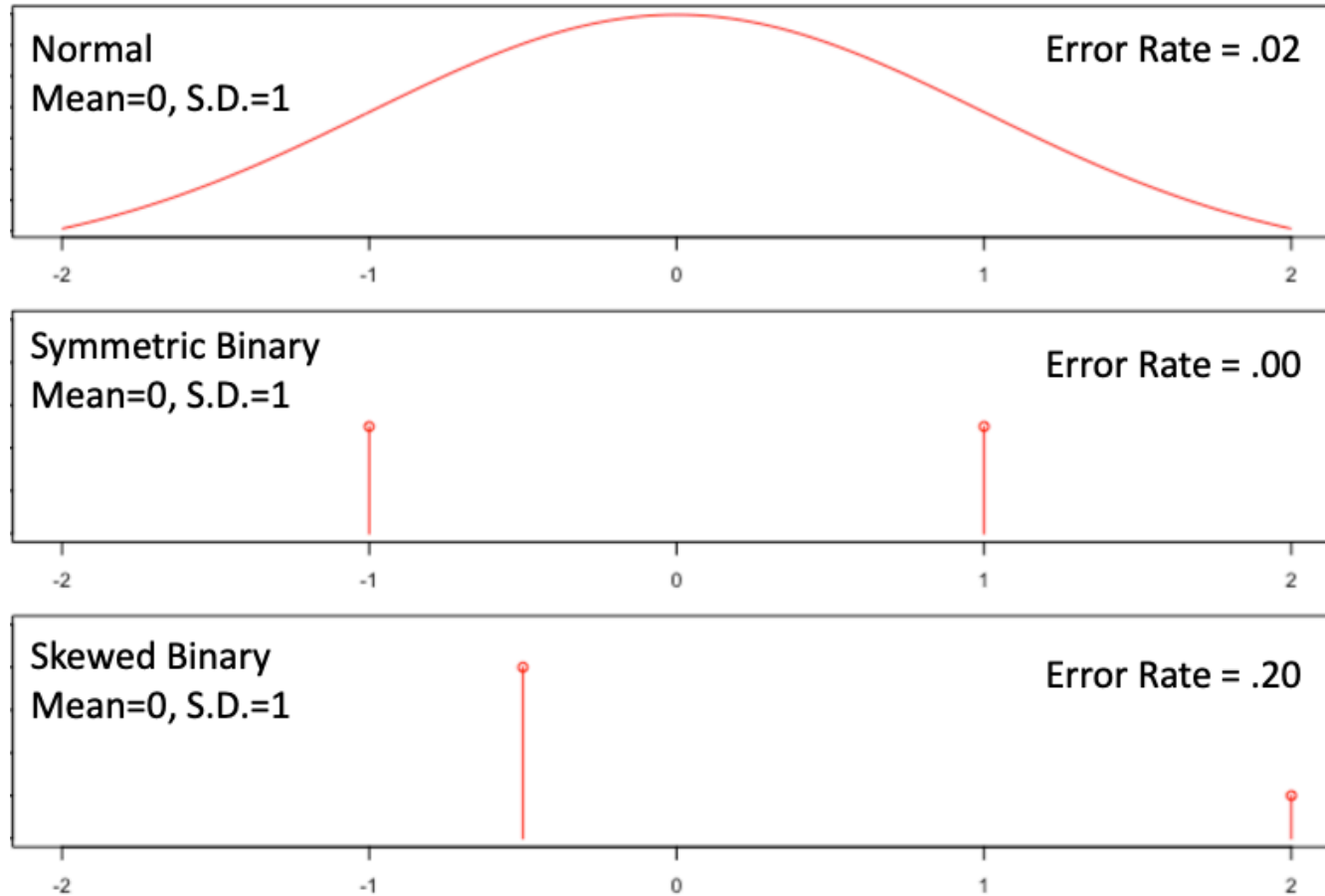


- Hidden fact: Population mean = 0
- Idea: Market if estimate is 2 standard errors above 0.

Speaker notes

- Here's an example. You want to understand whether vitamin W improves performance on a statistics test.
- You don't know this, but vitamin W doesn't do anything
- You estimate the performance, and it's positive, great!
- But you know that there's uncertainty. So you're not convinced the effect is really positive
- You decide on this rule: if the estimate is at least 2 standard errors away from zero, you'll take the vitamin to market.
- That's actually not a terrible rule of thumb. This is a very very simplified version of a hypothesis test , but I think there's enough detail to make this point.

The Importance of Shape



Speaker notes

- Here are three possible sampling distributions for the statistic. All 3 of them have a mean of 0 and a standard error of 1.
- So if you get an estimate of 2, you will take vitamin W to market. and that's a mistake because the true effect is zero.
- The question is, how likely are you to make a mistake in each case?
- So the shape really matters when we want to express uncertainty more precisely. That seems like a problem, because there are so many possible shapes out there.

The Central Limit Theorem

Central limit theorem (CLT): idea

For a __very broad__ class of population distributions, the sampling distribution of the mean becomes approximately normal as the sample size grows large.

Speaker notes

- It turns out that a very famous theorem can help us. It's called the CLT.
- The CLT says that, yes, there are a lot of possible distributions out there. If you have a small sample size, the variety is a big problem
- But as the sample size grows large, the sampling distribution of the sample mean becomes more and more normal.
- This is great because it's going to unlock a lot of important tools, including confidence interval and hypothesis tests.
- Because of that, this is often viewed as the most important theorem in all of classical statistics

Reading: The Central Limit Theorem

Reading: The Central Limit Theorem

Note: This is a reading call, we're just placing it here for organization.

Read pages 108 and 109 of section 3.2.4. There's no *need* to read through the demonstration of Slutsky's theorem, but you can if you would like.

- Rather than proving the CLT, we are going to ask you to work through a short demonstration against data that we hope will convince you of the CLT's effectiveness.
- The book is terse in its presentation of when and how the CLT applies. We will fill that out when we come back together.

Apply the Central Limit Theorem

Apply the Central Limit Theorem

Note: This is a learnosity activity, just placing it here for organization.

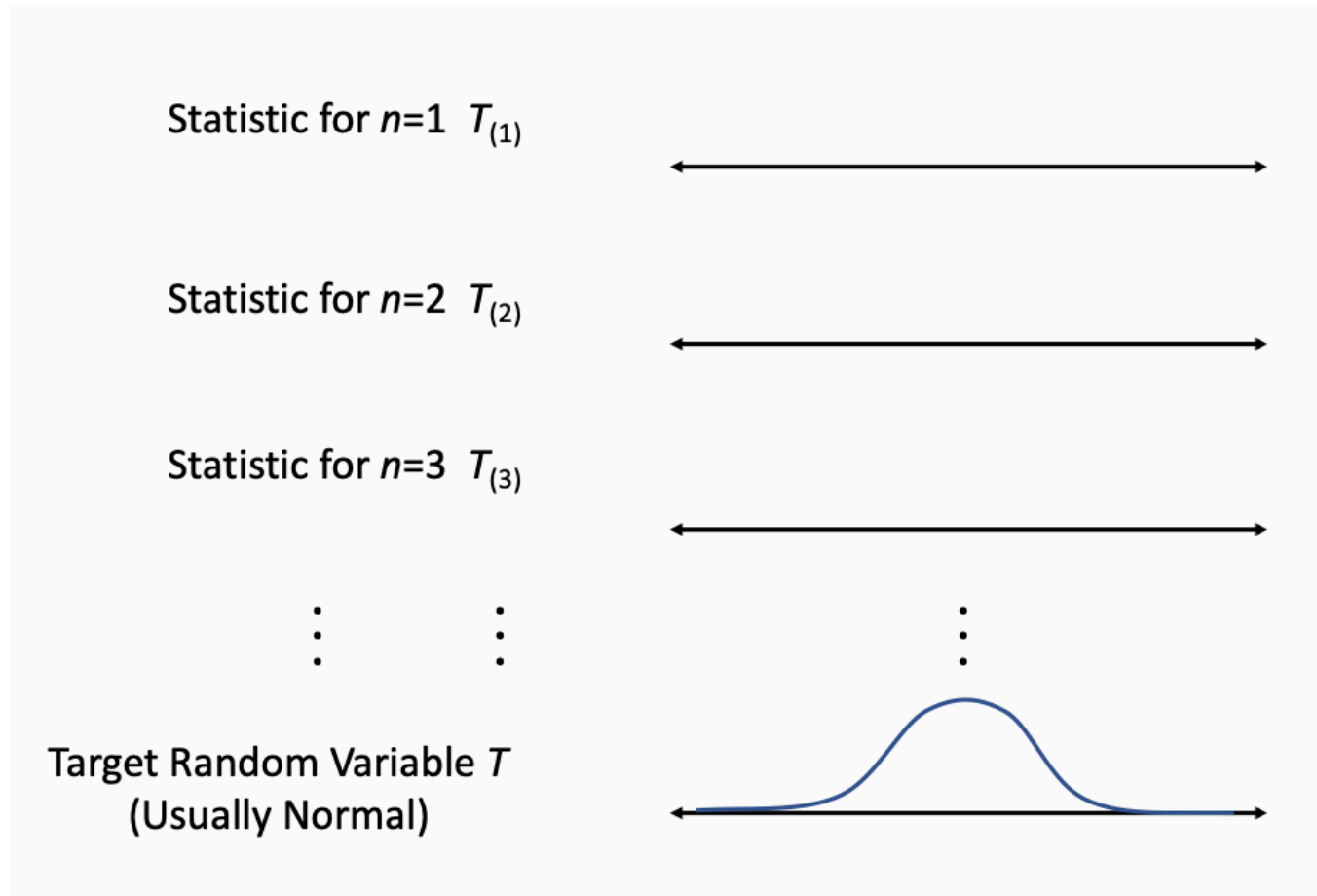
Central Limit Theorem

Reminder of Context

- The WLLN tells us what happens to the sample mean as $n \rightarrow \infty$: $\overline{X} \xrightarrow{p} \textcolor{red}{E}[X]$
- We also know that, as $n \rightarrow \infty$, we can generate an increasingly good estimate for $\textcolor{red}{V}[X]$ because $\overline{X^2} - \overline{X}^2 \xrightarrow{p} \textcolor{red}{V}[X]$
- For more precise statements about uncertainty, we need the sampling distribution of the statistic.

- We were really glad when we saw the result of the WLLN.
- From weak assumptions, we can put together an estimator
- What if we want to know more information about this estimator?

Convergence in Distribution



Speaker notes

- Let's build out the idea of
- Suppose you have a sequence of random variables, for now, it will be a set of sample means.
- First, suppose that you take a sample of size 1, and return the average. What will you see?
- Then, suppose that you take a sample of size 2, and return the average. What will you see? Will this larger sample have a larger or a smaller sampling variance of the mean? (This is the WLLN claim).
- The idea of convergence in distribution, is that as you go along the sequence of distributions, the shape
- Convergence in distribution is much stronger than just a single point converging like we had for the WLLN.
- The whole distribution not just a central tendency for example, converges to the mean

Convergence in Distribution

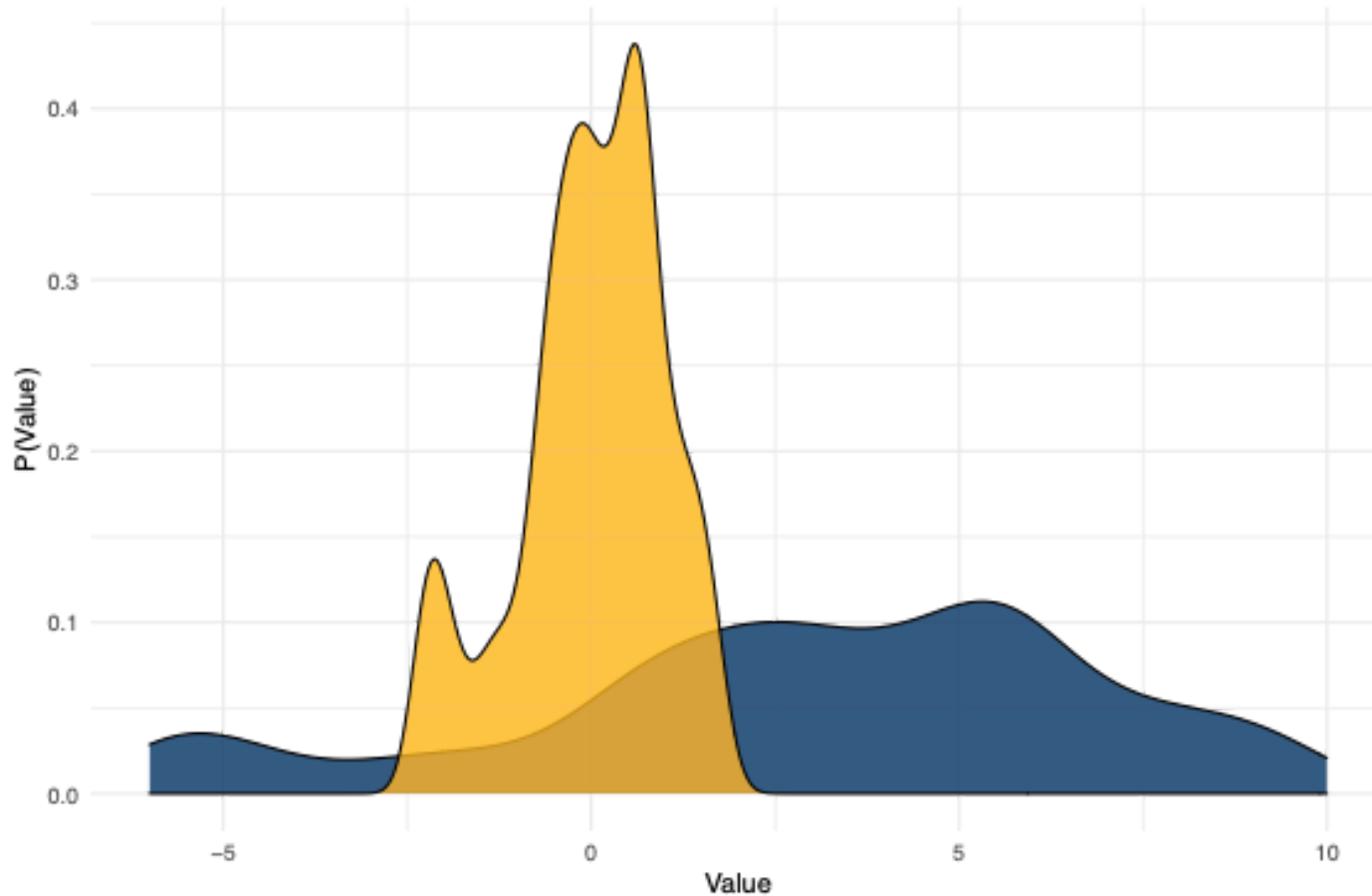
i Definition: Convergence in distribution

Let $(T_{(1)}, T_{(2)}, T_{(3)}, \dots)$ be a sequence of random variables, with cdfs $(F_{(1)}, F_{(2)}, F_{(3)}, \dots)$, and let T be a random variable with cdf F . Then $T_{(n)}$ *converges in distribution* to T if, for all $t \in \mathbb{R}$ at which F is continuous, $\lim_{n \rightarrow \infty} F_{(n)}(t) = F(t)$. We denote this as $T_{(n)} \xrightarrow{d} T$.

Speaker notes

- Here's the statement, and the thing to notice is that the actual statement of convergence is in terms of the CDFs
- You might hope it was about pdfs, but that's not possible. for example, if you have a discrete population, you can never have a continuous statistics
- Also, every random variable in the world has a cdf, so that makes this more universal.

The need to Standardize



Speaker notes

- We want to say that the distribution of the sample mean converges in distribution to the normal, but there's one more wrinkle.
- Remember that as n increases, the distribution gets thinner and thinner.
- The variance decreases as $1/n$
- So we can't just look at the raw distribution, which approaches a spike. We have to stretch it back out so that the width is a constant.

- Recall,

Standardizing the Sample Mean

Definition: Standardized sample mean

For IID random variables $(X_1, X_2, \dots, X_n)$ with finite $E[X] = \mu$ and finite $V[X] = \sigma^2$, then __the standardized sample mean__ is:

$$Z = \frac{\left(\overline{X} - E[\overline{X}]\right)}{\sigma \sqrt{n}} = \frac{\overline{X} - \mu}{\sigma \sqrt{n}}$$

- The standardized sample mean always has mean 0 and standard deviation 1.

The Central Limit Theorem

The Central Limit Theorem

Let $(X_1, X_2, X_3, \dots)$ be a sequence of i.i.d. random variables with finite mean $E[X] = \mu$ and finite variance $V[X] = \sigma^2$,
 $Z = \frac{\sqrt{n}(\overline{X} - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$ which is the equivalent to $\sqrt{n}(\overline{X} - \mu) \xrightarrow{d} N(0, \sigma^2)$

Speaker notes

- With all that out of the way, we can finally state the CLT
- To summarize: As long as n is large enough, then the mean,
- It doesn't matter what the population distribution looks like.
- This means that we can run hypothesis tests and make precise statements about uncertainty.

Generality of the CLT

The CLT applies to:

- Continuous Random Variables
- Discrete Random Variables
- Symmetric Random Variables
- Asymmetric Random Variables

Only Requirements:

- Data points are IID.
- Population has finite variance.

Versions of the CLT exist for many other statistics.

What Sample Size is Enough?

The CLT works in the limit as

$n \rightarrow \infty$. What can we say for a finite $n < \infty$?

- Rule of Thumb: $n = 30$ for CLT to “kick in.”
- Reality: Convergence depends on how non-normal population is. - A normal population requires $n = 1$.
- Highly skewed distributions may require $n = 100$, $n = 1000$, or more.

- When you were working with a normal underlying distribution, how quickly with a sample average move
- If two distributions have the same
- There's a general rule of thumb that for

Learnosity: When Does the CLT Apply?

Where and When Does the CLT Apply?

Part III

Which of these random variables has an approximately normal distribution because of the CLT? - X = hours spent by one individual on a 203 homework assignment - $\bar{X}$ = average hours spent by randomly assigned study groups of size 4 on the same 203 homework - X = number of barks by a dog named Rex for a one-hour period at night

Where and When Does the CLT Apply?

Part III

Which of these random variables has an approximately normal distribution because of the CLT?

- X = total number of barks by a neighborhood of dogs for a one-hour period at night
- X = age of a randomly selected MIDS student
- $\bar{X}$ = average of sample of 10 randomly selected MIDS students
- X = VADER sentiment of a sample of 100 SMS messages

The Plug-In Principle

The Plug-In Principle

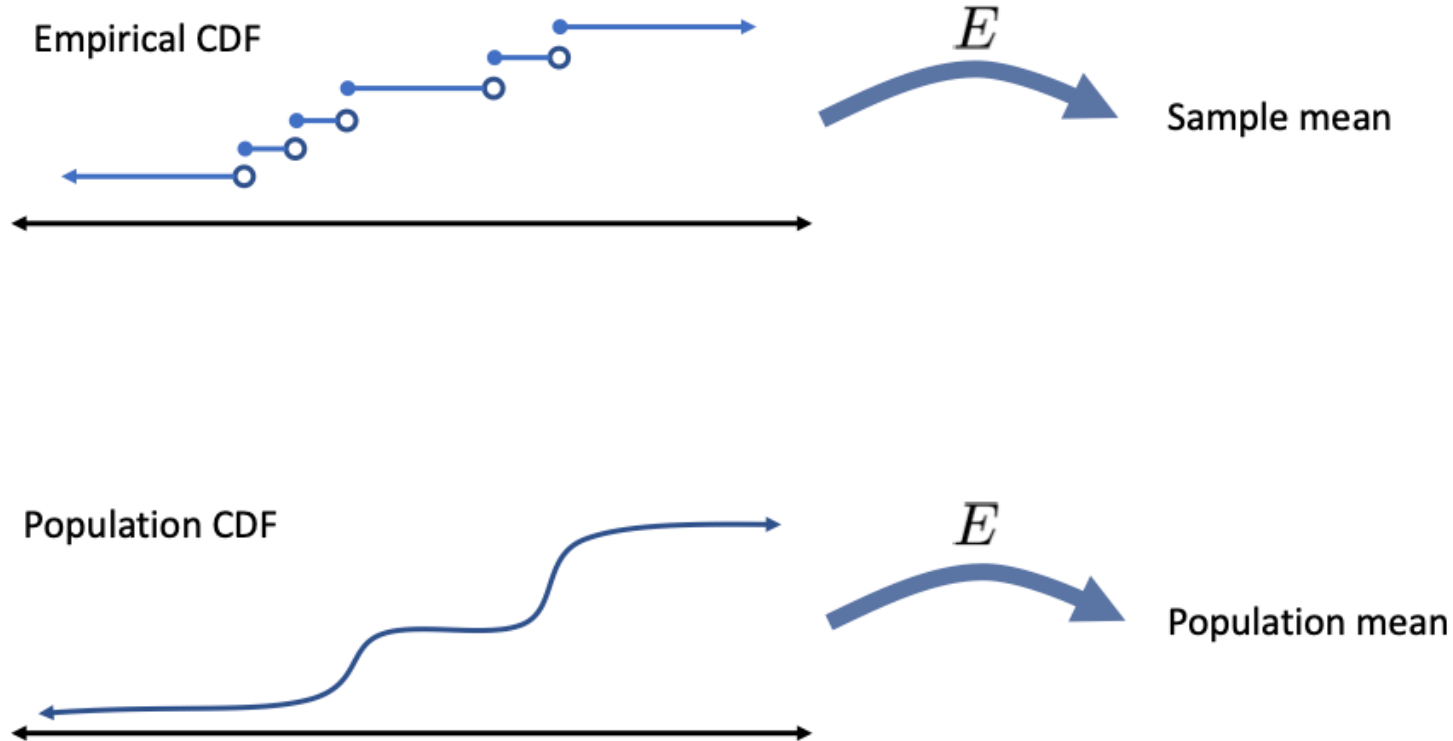
- Want mean of population $\rightarrow$ Mean of sample.
- Want variance of population $\rightarrow$ Variance of sample.
- Want $f(E[X], V[X])$? $\rightarrow f(\bar{X}, \hat{V}(X))$

See the Pattern?

Speaker notes

- I want to discuss this idea of the Plug-In principle
- The book uses it a lot, but it also glosses over the details
- Now you've seen a few estimators, and you may start to notice a pattern...
- If you want a function of the population variance, you can plug the sample variance into the function
- So statisticians started to notice, hey, if we just plug in these sample analogues, we seem to get good estimators.
- Is this just an accident? Or is there something deeper going on? Can we use this to find estimators for more parameters?

The Plug-In Principle



Speaker notes

- It turns out, yes, there is something going on
- At the bottom, we have a population, represented as a CDF. We apply expectation to it, to get the population mean
- Now at the top, we have our sample of data, represented by the empirical CDF. Well, if you take the empirical distribution and apply expectation to it, you get the sample mean.
- What's key here, is that these expectations are exactly the same operator! You really need to take measure theory to prove this, but hopefully that seems believable
- Now, what happens when
- We know the empirical CDF converges in distribution on the population CDF.
- We need the expectation operator to be continuous, and that's not easy to define here. but if it's smooth enough, as the empirical cdf approaches the

Reading Assignment

Reading Assignment

Only if you're interested, read section 3.3 - 3.3.1 to learn more about the Plug-In Principle.

Asymptotic Theory

Across a range of applications, small samples are problems. -

If we've got a *lot* of data, though, we can rely on weaker requirements - Asymptotic properties of estimators ask the question, **What happens when we have a lot of data?** - In general, as $n \rightarrow \infty$, does $\hat{\theta}$ converge in distribution or probability to something that is useful or desirable?

Speaker notes

- Tie the problems back to the small-sample bias of the Plug-In Sampling Variance

Reading: Asymptotics

Note: This is a **READING CALL**. We're placing it here for **organization**. - The interested student can read pages 111-114.
- But, we might recommend skipping it if you're constrained. -
The take home is that there is an asymptotic statement of: -
Being Normally Distributed - SE, MSE and Efficiency - Sampling
Variance and Sampling Standard Error

Reading: The Plug-In Principle

- $F(x)$ is the cumulative distribution function, the *CDF*
- $f(x)$ is the probability distribution function, the *PDF*
- $F(\hat{x}) \neq F(x)$ and $f(\hat{x}) \neq f(x)$
- As $n \rightarrow \infty$ we hope that $F(\hat{x}) \xrightarrow{d} F(x)$ and $f(\hat{x}) \xrightarrow{d} f(x)$.

- We're not going to lie to you... this is a challenging section to read.
- A couple of notes are in order.

The Plug-In Principle

Expectation and Variance Functionals

You _know_ what the processes for calculating an expectation and variance are.

```
\begin{align*}
```

$$E[X] \triangleq T_{\{E\}}(F) = \int x \cdot dF(x) \quad \backslash \backslash$$

$$V[E] \triangleq T_{\{V\}}(F) = \int (x - E[X])^2 \cdot dF(x)$$

```
\end{align*}
```

These are the *statistical functionals* for expectation and variance.

Speaker notes

- You might be annoyed that we're dealing with
-

The Plug-In Principle (cont.)

Plug-In Estimators

For i.i.d. random variables $X_1, X_2, \dots, X_n$, with a common CDF F , the plug in estimator for $\theta = T(F)$ is just $\hat{\theta} = T(\hat{F})$.

$\begin{aligned} \hat{E}(X) &= T_E(\hat{F}) \quad \backslash \backslash \\ &= \sum x \cdot \hat{f}(x) \quad \backslash \backslash \\ &= \sum x \cdot \frac{I(X_i = x)}{n} \quad \backslash \backslash \\ &= \frac{1}{n} \sum x \quad \backslash \backslash \\ &= \overline{X} \quad \backslash \backslash \end{aligned}$

$\end{aligned}$

Example of the Plug-In Principle

Example with Discrete Data

Suppose you there is some discrete process that places values into the random variable X .

`\note[item]{(You can't see equation process!)}`

- $F_X(1) = 1/3$
- $F_X(2) = 1/6$
- $F_X(3) = 0$
- $F_X(4) = 1/2$

But, you do get to produce i.i.d. random draws from the process. Of 1000 draws: `\\begin{tabular}{llll}`

`\textbf{1} & \textbf{2} & \textbf{3} & \textbf{4}`

`\hline`

`331 & 156 & 0 & 513`

`\end{tabular}`

Example of the Plug-In Principle, cont'd

i Example with Discrete Data

$$E(\text{bs}X) = T_E(\hat{F})$$

Handwritten calculations for the Plug-In Principle with discrete data:

Given data: $(1, 2, 3, 4)$ with frequencies $(33, 156, 0, 513)$.

Step 1: $\hat{F} = T(\hat{F})$

Step 2: $E[X] = T_E[\hat{F}]$

Step 3: $E[X] = \int_{-\infty}^{\infty} x \cdot d\hat{F}(x)$

Step 4: $E[X] = \sum_x x \cdot \hat{F}(x)$

Step 5: $E[X] = \sum_x x \cdot \hat{f}(x)$ (this is the table above)

Step 6: $E[X] = \frac{1}{4} \sum x \cdot f(x)$

Step 7: $E[X] = \left(1 \cdot \frac{33}{1000}\right) + \left(2 \cdot \frac{156}{1000}\right) + 0 + \left(4 \cdot \frac{513}{1000}\right)$

Step 8: $E[X] = \frac{331}{1000} + \frac{312}{1000} + \frac{2052}{1000}$

Step 9: $E[X] = \frac{2695}{1000} = 2.695$

Step 10: $E[\hat{X}] = 2.695$

Step 11: $E[\hat{X}] = 2.66$

Reading: Kernel Methods Estimate the PDF

Read pages 121 - 124.

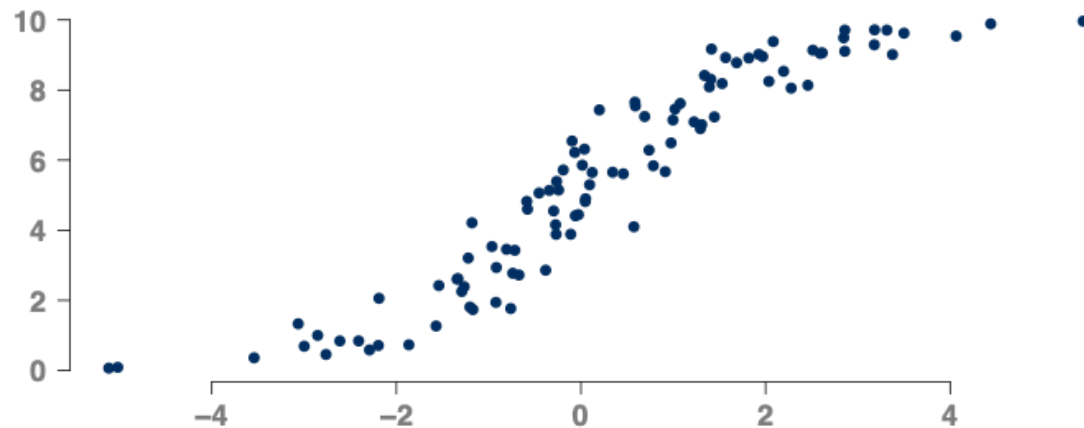
Kernel Methods, Part I

Kernel Density Estimator of PDF

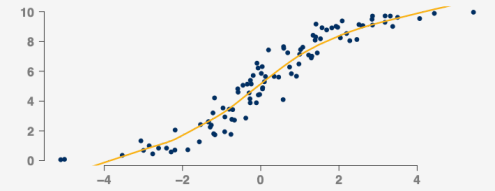
- Cannot *directly* observe the joint PDF.
- For discrete RV, approximations come through frequency tables.
- For continuous RV, approximations come through kernel density estimates: $\hat{f}_n(x) = \frac{1}{n} \sum_{i=1}^n K_h(x - X_i)$

Kernel Methods, Part II

- Kernel methods are smoothing methods
- The *kernel* is some weighting function



Speaker notes



Kernel Methods, part III

As smoothing function, these permit plug in estimates that are directly analogous to the estimating functionals of the CDF.

Kernel Plug-In Estimator

The _Kernel plug-in estimator_ of $\theta = T(F)$ is $\hat{\theta}_K = T(\hat{F}_K)$ where, $\hat{F}_K = \int_{-\infty}^x \hat{f}(u) du$.

Kernel Methods, part IV

Kernel Plug-In Estimator for Expected Value

We know the functional for the expected value has a specific
 ``shape'' $E[Y] = \int y \cdot dF(y)$. So, the feasible kernel method is
 $\hat{E}_k(Y) = \int_{-\infty}^{\infty} y \cdot d\hat{F}_K(y)$

Kernel Methods, part V

- $\hat{\mathbb{E}}[Y]$ and $\hat{\mathbb{E}}[Y|X]$ are lowest MSE estimates of Y

$$\hat{\mathbb{E}}_K(\text{abs } Y) = \int y f_{\hat{Y}}(y)$$

$$\hat{\mathbb{E}}_K(\text{abs } Y | X = x) = \int y f_{\hat{Y}|X}(y|x)$$

- If you can produce i.i.d. samples from $f_{Y|X}$, you can produce estimates, $f_{\hat{Y}|X}(y|x)$ that get ever closer to the true value

- The next sections in the course combine these facts with the CLT distributional shapes so that we can make

Apply Kernel Methods

Note: This is a LEARNOSITY activity. We're just placing it here for organization.